

Your grade: 96%

Your latest: 96% • Your highest: 96% • To pass you need at least 80%. We keep your highest score.

Next item →

coach

Almost there! Let's review your answers and then try again.

Your work showed a strong understanding of pretrained models, LSTM cells, the Adam optimizer, and tensor manipulation in PyTorch. However, there is room for improvement in understanding the benefits of transfer learning. Before retrying this assignment, we recommend focusing on this key area:

- **Transfer Learning and Pre-Trained Networks (101):** In Question 2, your understanding of the benefits of transfer learning was incomplete. Try reviewing [this item](#) to refresh your memory.

Good job on the rest of the assignment!

Retry assignment



1. What is the primary purpose of using a pretrained model in neural network training?

1 / 1 point

- ☐ To prevent overfitting by using a smaller dataset.
- ☐ To ensure 100% accuracy in classification tasks.
- ☒ To save time and computational resources by leveraging already learned features.
- ☐ To avoid the need for a validation dataset.

✔ Correct

Correct! Pretrained models help save time and computational resources by using already learned features.

2. Which of the following are benefits of using transfer learning in neural networks?

0.8 / 1 point

- ☒ Reduced training time.

✔ Correct

Correct! Transfer learning reduces training time by leveraging pre-learned features.

- ☐ Enhanced feature extraction.

- ☒ Improved model accuracy with less data.

✔ Correct

Correct! Transfer learning can improve accuracy even with less data by utilizing features from pretrained models.

- ☐ Simplified model architecture.

- ☐ Elimination of the need for data augmentation.

You didn't select all the correct answers

3. What is an advantage of Long Short-Term Memory (LSTM) cells over basic Recurrent Neural Network (RNN) cells?

1 / 1 point

- ☒ LSTM cells can better capture long-term dependencies in data
- ☐ LSTM cells require less computational power
- ☐ LSTM cells are simpler to implement
- ☐ LSTM cells eliminate the need for activation functions

✔ Correct

Correct! LSTM cells are specifically designed to address the vanishing gradient problem, making them more effective at capturing long-term dependencies.

4. What is the primary purpose of using the Adam optimizer in training an LSTM model?

1 / 1 point

- ☐ To reduce the number of training epochs required
- ☐ To maintain a constant learning rate
- ☐ To increase the model's complexity
- ☒ To adapt the learning rate for each parameter

✔ Correct

Correct! The Adam optimizer adjusts the learning rate for each parameter, which helps in faster and more efficient convergence.

5. Which PyTorch function is used to change the shape of a tensor without changing its data?

1 / 1 point

- ☒ view
- ☐ squeeze
- ☐ expand
- ☐ reshape

✔ Correct

Correct! The `view` function in PyTorch is used to reshape a tensor without altering its data.